

Scribing Note: The foundation of modern biology and genomics: Sequence data

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BMEG3105
16 September 2026

Outline of Lecture 3:

- Biological Sequence Data
- Sequencing Technologies (DNA, RNA, and Protein)
- Sequence Comparison, Alignment, and Similarity
- Enumeration
- Dynamic Programming (DP)
- Traceback

Biological Sequence Data:

Central Dogma: Genetic information is stored inside DNA sequences, transcribed into RNA, and translated into protein products.

$$\text{Phenotype} = \text{Genotype} + \text{Environment}$$

(genotype encoded in sequences)

Main Types of Sequences:

DNA: Composed of A, T, C, G, forming a complementary double helix ($\approx$ 3 billion base pairs in the human genome)

RNA Sequence: Composed of A, U, C, G

Protein Sequence: Built from chains composed of 20 standard amino acids

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Sequencing Technologies:

Nanopore Sequencing: A technique in which a DNA strand passes through a chemical pore. The distinct nucleotide bases “create” electrical current in different ways. The identity of each base is determined by measuring variations in the that current in real time.

Protein Sequencing (Mass Spectrometry MS): Long sequence chains are divided into short pieces. Then MS determine the weight of each pieces. Researchers then assemble these fragments back into full-length sequences like assembling a jigsaw puzzle.

DNA Analysis Pipeline (After obtaining DNA sequences)

1. Perform quality control to assess and filter raw sequencing reads.
2. Map reads to a reference genome.
3. Perform variant calling to identify differences from the reference.
4. Associate variants with phenotypes

Sequence Comparison, Alignment, and Similarity

Similarity:

Protein sequences: similar sequences may imply similar structures and therefore similar functions. This is called the sequence-to-structure-to-function paradigm.

High similarity can also imply a common evolutionary ancestry(homology) which may help researchers study important biological mechanisms.

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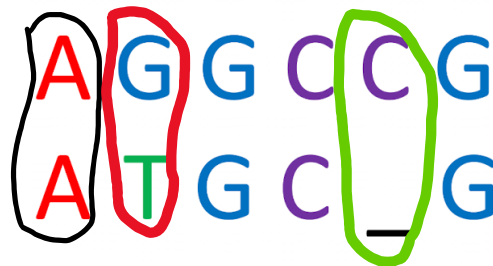
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Alignment:

Pairwise sequence alignment arranges two sequences to maximize their similarity degree. Gaps can be inserted into either sequence when useful.

Method

◆ Match: the same symbol is aligned, for example A with A.



● Mismatch: different symbols are aligned, for example G with T

● Gap: a symbol is aligned with a gap, for example C with nothing

Alignment Scoring:

Scoring matrix:

	A	C	G	T
A	2	-7	-5	-7
C	-7	2	-7	-5
G	-5	-7	2	-7
T	-7	-5	-7	2

Gap penalty = -10

$$\text{Alignment score 1} = 2 + (-7) + 2 + 2 + (-10) + 2 = -9$$

A G G C C G
A T G C _ G
 ↑
 choosing gap

$$\text{Alignment score 2} = 2 + (-7) + 2 + 2 + (-7) + (-10) = -18$$

A G G C C G
A T G C G _
 ↑
 choosing gap

The upper one is more “possible” because the alignment score is larger than the bottom one.

Enumeration

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1. Enumerate all possible alignments of 2 sequences
2. Calculate the alignment score of each possibility
3. Find the alignment that have the highest score



“False” method

Why?

The number of possible alignments grows extremely quickly as sequence length (n) increases

$$\binom{2n}{n} = \frac{(2n)!}{n!n!}$$

When $n = 300$, this is approximately 7×10^{88} , which is far too large to search exhaustively.

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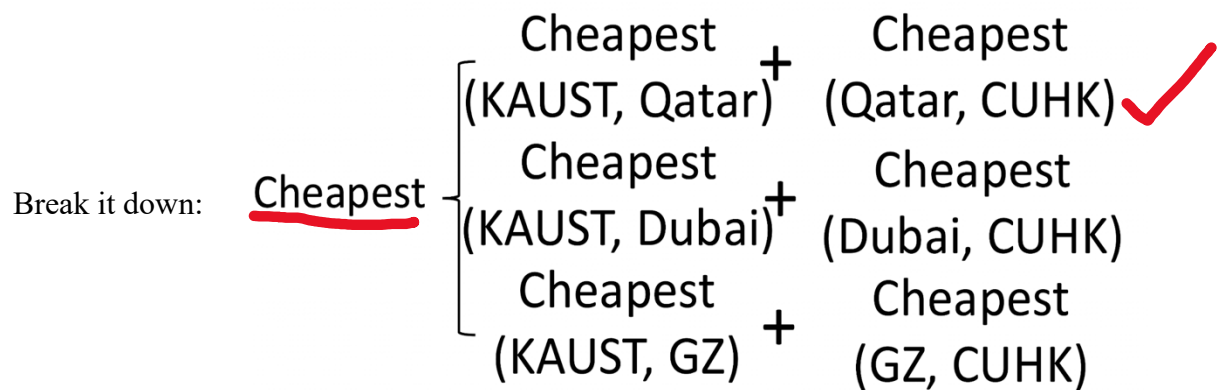
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Dynamic Programming (DP)

Dynamic programming solves large problems by breaking them down into smaller subproblems. These subproblems are then solved optimally and recursively, and their results are combined to construct for the original large problem.

e.g large Problem: Cheapest flight from KAUST to CUHK



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1. draw a $(i+1) \times (j+1)$ table
eg. ACCG & ACG (3+1) x (4+1)

$$F(i, j) = \text{Best} \begin{cases} F(i-1, j) + \text{gap} \\ F(i, j-1) + \text{gap} \\ F(i-1, j-1) + SC(x_i, y_j) \end{cases}$$

	gap	A	C	C	G
gap	0, 0	0, 1	0, 2	0, 3	0, 4
A	1, 0				
C	2, 0				
G	3, 0				

First row & column:

from 0,0 go to 0,4 and 3,0, each step add -10
since we add a gap, with gap penalty -10



	gap	A	C	C	G
gap	0	-10	-20	-30	-40
A	-10	2	1, 2		
C	-20	2, 1	2, 2		
G	-30				

$$F(A, A) = (1, 1) = SC(A, A) = 2$$

$$F(A, C) = (1, 2)$$

3 possibility $\left\{ \begin{array}{l} -20 \text{ add a gap} = -20 + (-10) = -30 \text{ (down)} \\ 2 \text{ add a gap} = 2 + (-10) = -8 \rightarrow \text{largest score } -8 \text{ (right)} \\ -10 \text{ add } SC(A, C) = -10 + (-7) = -17 \text{ (diagonal)} \end{array} \right.$

$$\therefore (1, 2) = -8$$

do same thing in 2, 1 & 2, 2

$$(2, 1) = -8 \quad ; \quad (2, 2) = 4$$

	gap	A	C	C	G
gap	0	-10	-20	-30	-40
A	-10	2	-8		
C	-20	-8	4		
G	-30				



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Repeat the step to fill the remaining cell

	gap	A	C	C	G
gap	0	-10	-20	-30	-40
A	-10	2	-8	-18	-28
C	-20	-8	4	-6	-16
G	-30	-18	-6	-3	-4

$$F(A, C, C) = (2, 3) \left\{ \begin{array}{l} -18 + \text{gap} = -18 + (-10) = -28 \quad (\text{down}) \\ 4 + \text{gap} = 4 + (-10) = -6 \quad (\text{right}) \\ -8 + S(C, C) = -8 + 2 = -6 \quad (\text{diagonal}) \end{array} \right\} \text{largest score}$$

there are 2 route to "arrive" -6

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		A	C	C	G
	0	-10	-20	-30	-40
A	-10	2	-8	-18	-28
C	-20	-8	4	-6	-16
G	-30	-18	-6	-3	-4

①

Optimal alignment 1

ACCG
A _ CG

②

Optimal alignment 2

ACCG
AC _ G

so there are 2 "arrow" meaning may have 2 optimal alignment

∴ the "best route" really from -6 to -4

$$= -6 + S(G, G)$$

$$= -6 + 2$$

$$= -4$$

∴ 2 Optimal alignment

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Traceback

Storing an arrow in each DP-table cell records the cell that produced its optimal value. Starting from the bottom-right cell and following the arrows back to the top-left cell reveals the optimal alignments, not just the score.